

Your Dream Pet

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March 15, 2026

1. Abstract

This project examines how structured data analysis can improve the accessibility of pet adoption information on online rescue platforms. Data for 500 adoptable pets in California were collected from RescueMe.org using web scraping. The listings were organized into a dataset containing location, breed, compatibility traits, temperaments, and health indicators. Visual analysis of this dataset shows that rescue pressure is concentrated in a few counties, particularly Riverside County, and that many animals display balanced temperament and compatibility traits suitable for typical households. Text analysis of pet descriptions and an ordered logistic regression model using the PetFinder dataset further suggests that richer descriptions and a greater number of photos are associated with faster adoption outcomes. Overall, these findings highlight how better organization of adoption data can make pets easier to discover and help shelters allocate resources more effectively.

2. Introduction

Animal shelters across the United States often operate with limited funding and rely on volunteers to manage daily operations (ASPCA, 2023). As a result, information about adoptable pets is often stored in inconsistent formats, making it difficult for potential adopters to search for animals. Improving how this information is organized could make the adoption process more efficient by connecting animals with suitable homes more quickly.

Data for this study were collected from RescueMe.org, a website that lists adoptable pets from shelters and rescue organizations across the United States. Although each animal profile includes information such as personality traits, energy levels, and compatibility with children and other animals—traits that often influence adopter decisions (Weiss et al., 2012), the website only allows users to filter listings by the sex of the animal.

To address this, the listing information was scraped and organized into a structured pandas DataFrame. Visualization techniques were used to explore patterns in the dataset, and natural language processing methods were applied to analyze the textual descriptions of pets on adoption speed. The goal of the project is to show how structured data and text analysis can improve the ability to search for pets that match adopter preferences.

3. Methods

3.1 Data Acquisition

Data was collected from RescueMe.org, a website that lists adoptable pets across the United States. Each pet profile contains structured information such as name, breed, age, sex compatibility traits, energy

levels, health information, and county location. It also includes unstructured textual descriptions of the animals.

A web scraping pipeline was developed in Python to extract these attributes from individual pet profile pages. Custom getter functions were created to retrieve specific fields from the HTML content. The getter functions were then combined into a larger scraper function that iterated through pet listing URLs and collected the relevant information. The extracted data were stored in a pandas DataFrame for further analysis.

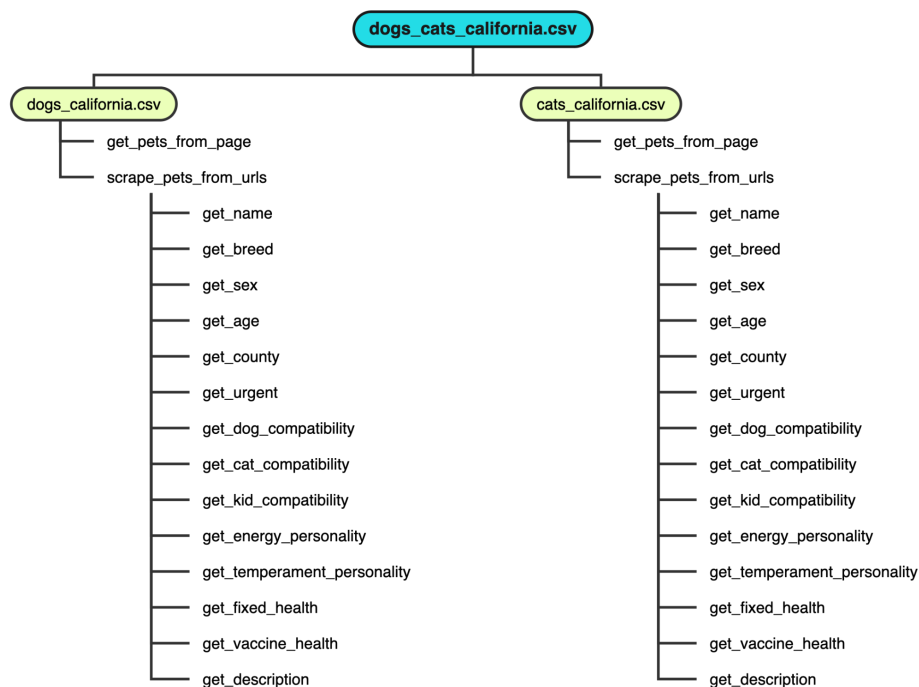


Figure 1: Visual Representation of the Web Scraping Process

3.2 Data Cleaning

Throughout the web scraping process, the data scraped required further cleaning and processing to ensure consistency and usability for analysis.

3.2.1 Missing Values

Some pet profiles contained incomplete information, such as no name, breed, age, sex, or certain characteristics. Instead of dropping these profiles from our data frame and causing pets with incomplete profiles to lose exposure, we listed the missing information as “Unknown”. This ensures that our final dataset was transparent about the amount of information available on each pet without having to skip any pet’s profile.

3.2.2 Transformation

While most of the pet data was able to be gathered by simply scraping the exact text from each pet’s profile, five characteristics—compatibility, energy, temperament, sterilization status, and vaccination

status—involved a more in-depth process. The information regarding these traits are contained in longer, more verbose bullet points on each pet’s page—for example, “Good with Most Dogs”, “Low Energy”, “Very Submissive”, “Neutered”, or “Needs to be Vaccinated”. Thus, these traits needed to be transformed into quantifiable categorical variables to allow easier filtering and analysis.

For these traits, we analyzed numerous profiles to determine the key words included in each bullet point. We discovered that bullet points regarding compatibility always contained the key words “Good” or “Not Good” in addition to the type of animal or person, bullet points regarding energy contained “Low”, “Average”, or “High”, bullet points regarding temperament contained “Submissive”, “Average”, or “Dominant”, bullet points regarding sterilization contained simply “Spayed” or “Neutered” or had “Not” preceding the appropriate operation, and bullet points regarding vaccination contained “Current” or the preceding word “Needs”.

By scraping for these key words and simplifying their content to be more easily read and understood at a glance, we hoped to make the adoption application process more efficient and approachable for prospective adopters.

3.2.3 Standardization

To ensure a consistent format for each column of our data frame, we applied `str.strip()` to all data scraped to eliminate leading or trailing whitespaces. For the visualization process, we further stripped the data to ensure that a value like "Orange County " was reduced to "Orange," matching the official keys in our GeoJSON file.

3.3 Feature Engineering

3.3.1 Urgent Status Extraction from Text

Depending only on official labels may result in significant missing data. Our analysis has found that there are serious inconsistencies in how urgency status is reported on rescueme.org website. While some animals in urgent adoption need to have official labels, there are also cases where emergency information is simply entered manually into the ‘Name’ field like ‘Urgent on Feb 26th’. These animals are overlooked by the platform’s official filtering system.

Therefore, we developed an identification process grounded in Natural Language Processing and Pattern Matching. We initiated the analysis with an exploratory text mining of pet names, which revealed high-frequency unigrams like "urgent" and bigrams like "after 03/10." Then we designed a regular expression pattern (`all_pattern`) to systematically extract warning signals, including high-risk keywords and specific temporal markers such as date formats and monthly abbreviations. These extracted signals were then integrated into a `categorize_urgency_nlp` function that cross-references official labels with keywords to create an advanced urgency column. This approach ensures that high-risk animals previously overlooked by the system are now correctly identified and prioritized for rescue resources.

3.3.2 Simplification of Breed Categories

The "breed" variable contained over 100 unique categories, leading to the visualization results being chaotic. We implemented a "Top 10 + Others" grouping strategy. Breeds outside the top 10 in frequency

were aggregated into a single "Other Breeds" category. This step significantly improved the signal-to-noise ratio in our visualizations, allowing us to identify dominant breed trends without losing the context of the overall population diversity.

3.3.3 Multi-Level Compatibility Tiers

To better align pet profiles with real-world adopter needs, we designed a tiered compatibility framework that transforms binary indicators into a multi-level Adoption model. This model categorizes pets into three distinct levels of adoption: a base tier for compatibility with children, a secondary tier for those compatible with both children and dogs, and a premium "Triple Friendly" tier. This hierarchical approach allows us to quantify the adaptability of the shelter population, providing a precision matching to identify animals that can integrate into complex households.

3.4 Visualization Strategy

Our choice of visualization libraries and styling logic was considered to maximize the clarity and authenticity of the data presentation.

3.4.1 Library Selection

For geospatial analysis, we chose Folium due to its support for interactive choropleth maps. It enables integration with GeoJSON data, allowing for exploration of the rescue pressure in each county of California. For the personality distribution, we chose Seaborn for the personality heatmap. We utilize its efficient processing of matrix data to identify the concentration of cross-variables. Finally, we chose Matplotlib for constructing donut charts, stacked bar charts, and grouped bars. It allowed for the precise manipulation of every graphical element and custom annotation. For example, we inserted the white circle in pie charts to reduce visual crowding.

3.4.2 Optimizing Visual Contrast

To preserve visual interpretability and data honesty in the face of skewness and outliers, we applied specialized color strategies tailored to our dataset's distribution. To address the right-skewed distribution of our data, we used a non-linear binning strategy for our maps. By manually defining `my_bins`, we ensured that the final visualizations objectively reflect the rescue pressure gradient rather than just highlighting the most extreme cases. In a similar way, we managed significant outliers in the personality heatmap through color capping. By setting the `vmax = 60`, we ensured that the extreme frequency of the central cell did not flatten the color variation of the remaining matrix.

3.5 Predictive Adoption Speed Modeling

To further investigate factors influencing pet adoption outcomes, we incorporated an external dataset from the PetFinder Adoption Prediction project. This dataset contains approximately 15,000 animals with recorded `AdoptionSpeed` values ranging from 0 to 4, where lower values correspond to faster adoption.

Initial correlation analysis suggested that structured attributes such as photo count, age, sterilization status, and vaccination status are associated with adoption outcomes. However, these variables individually explain only a limited portion of the variation in adoption speed. This observation motivated

the use of Natural Language Processing (NLP) techniques to extract additional information from the textual descriptions of pets.

3.5.1 Feature Extraction from Descriptions

Pet descriptions contain important contextual information about an animal’s personality, behavior, and health condition. To capture these signals, we implemented a text processing pipeline consisting of tokenization and part-of-speech tagging. Because descriptive adjectives often summarize key personality traits in adoption listings, we focused specifically on extracting adjectives from each description. To quantify the importance of these words across the dataset, we applied the Term Frequency–Inverse Document Frequency (TF-IDF) weighting scheme. For a term t in document d , the TF-IDF score is defined as

$$TFIDF(t, d) = TF(t, d) \cdot \log\left(\frac{N}{1+n_t}\right)$$

where

- $TF(t, d)$ is the frequency of term t in document d
- N is the total number of documents
- n_t is the number of documents containing term t

Using TF-IDF scores, we identified the top 20 most informative adjectives across the texts of pet descriptions. These adjectives were selected as candidate textual features. In addition, we computed the description length as a numerical feature representing the amount of information provided in each listing. Instead of using TF-IDF scores directly in the regression model, we constructed binary indicator variables representing whether each selected adjective appears in the description. The final feature vector therefore consists of the description length together with the presence indicators of the selected adjectives. In this project, TF-IDF features were computed using the `TfidfVectorizer` implementation from the `scikit-learn` library.

3.5.2 Ordered Logistic Regression Model

Because the `AdoptionSpeed` variable represents an ordered categorical outcome, we modeled it using an Ordered Logistic Regression framework. Let Y denote the adoption speed category and X represent the feature vector consisting of the description length and the binary indicators of the selected adjectives. The cumulative logit model is defined as

$$\log\left(\frac{P(Y \leq j)}{P(Y > j)}\right) = \theta_j - \beta^T X$$

for $j = 0, 1, 2, 3$, where

- θ_j are threshold parameters separating adjacent adoption categories
- β is the vector of regression coefficients
- X contains the NLP-derived features

The model estimates how textual characteristics influence the probability that a pet falls into faster or slower adoption categories. After training the model on the `PetFinder` dataset, the fitted model was applied to the California rescue dataset collected through web scraping earlier in the project. Using the same feature extraction procedure, predicted adoption speed scores were generated for each animal profile, allowing us to rank pets by their estimated adoption likelihood.

3.7 Technical Stack

The project was developed using Python 3 within a Jupyter Notebook environment. The following libraries were essential for the pipeline:

1. Data Acquisition: requests and lxml
2. Data Processing: pandas, numpy and re
3. Data Visualization: matplotlib, seaborn and folium
4. Natural Language Processing: nltk
5. Predictive Modeling: statsmodels and scikit-learn

4. Results

4.1 Data Frame

Name	Species	Breed	Sex	Age	County	Urgent Status	Dog Compatibility	Cat Compatibility	Kid Compatibility	Energy	Temperament	Spayed/Neutered	Vaccinated	Description	URL
PECK	Dog	Bull Terrier	Male	Young Adult	Riverside County	No	Unknown	Unknown	Unknown	Unknown	Unknown	Unknown	Unknown	Weight- 45lbs Peck	https://post.resc
MORGAN	Dog	Bull Terrier	Female	Young Adult	Riverside County	No	Unknown	Unknown	Unknown	Unknown	Unknown	Unknown	Unknown	Weight- 35lbs Morg	https://post.resc
Mack	Dog	German Sho	Male	Young Adult	San Bernardino C	No	Good	Good	Good	High	Average	No	Yes	Mack is a handsome	https://post.resc
Kingsley	Dog	Corgi	Male	Adult	Yolo County	No	Good	Unknown	Not Good	Average	Dominant	Yes	Yes	Kingsley is a 5 yr old	https://post.resc
Gwen	Dog	Great Dane	Female	Young Adult	San Joaquin Cou	No	Unknown	Good	Good	Average	Average	No	Unknown	is a good dog that ju	https://post.resc
URGENT AFT	Dog	French Bulld	Female	Young Adult	Riverside County	Yes	Unknown	Unknown	Unknown	Unknown	Unknown	No	Yes	ID#A549362 I am di	https://post.resc
URGENT AFT	Dog	Pit Bull mix	Male	Young Adult	Riverside County	Yes	Unknown	Unknown	Unknown	Unknown	Unknown	No	Yes	ID#A549364 I am di	https://post.resc
URGENT AFT	Dog	Chihuahua n	Female	Puppy	Riverside County	Yes	Unknown	Unknown	Unknown	Unknown	Unknown	No	Yes	ID#A549372 I am de	https://post.resc
URGENT AFT	Dog	Chihuahua n	Female	Young Adult	Riverside County	Yes	Unknown	Unknown	Unknown	Unknown	Unknown	No	Yes	ID#A549373 I am de	https://post.resc
URGENT AFT	Dog	Poodle mix	Female	Young Adult	Riverside County	Yes	Unknown	Unknown	Unknown	Unknown	Unknown	No	Yes	ID#A549385 I am di	https://post.resc
URGENT AFT	Dog	Basset Houn	Female	Puppy	Riverside County	Yes	Unknown	Unknown	Unknown	Unknown	Unknown	No	Yes	ID#A549386 I am di	https://post.resc
URGENT AFT	Dog	Lhasa Apso /	Male	Senior	Riverside County	Yes	Unknown	Unknown	Unknown	Unknown	Unknown	Yes	Yes	MOOSE - ID#A48778	https://post.resc
URGENT AFT	Dog	Cairn mix	Female	Unknown	Riverside County	Yes	Unknown	Unknown	Unknown	Unknown	Unknown	No	Yes	ID#A549374 I am de	https://post.resc

Table 1: Data Frame of Rescue Pet Attributes

Table 1 shows a sample of the structured dataset created from [RescueMe.org](https://www.rescueme.org) pet listings. Each row represents an adoptable animal, while the column contain attributes such as species, breed, location, compatibility traits, temperament, energy level, and health indicators including vaccination and sterilization status. Some entries appear as “Unknown”, reflecting missing information in the original listings. These records were retained to ensure all animals remained represented in the dataset. This structured data frame provides the foundation for the visual and predictive analysis presented in the following sections.

4.2 Visualization Analysis

The following visualizations quantify the current situation and provide data support for resource allocation.

4.2.1 Geospatial Concentration of Rescue Pressure

The Choropleth map (Figure 2) and the Top 10 Counties Bar Chart (Figure 3) are tools for identifying geographic hotspots. Our geospatial analysis shows that rescue pressure is heavily concentrated in Southern California. Specifically, Riverside County stands at the top with over 170 animals, which is substantially higher than the second-ranked Los Angeles County. The extreme outlier status of Riverside County indicates a potential crisis in local shelter capacity. It shows that while most counties have fewer than 10 pets, a few core regions are currently experiencing significant pressure on animal density. This finding is of high value for resource allocation, providing a clear reference for shelters and non-profits to direct funding and volunteer efforts where the pressure is most acute.

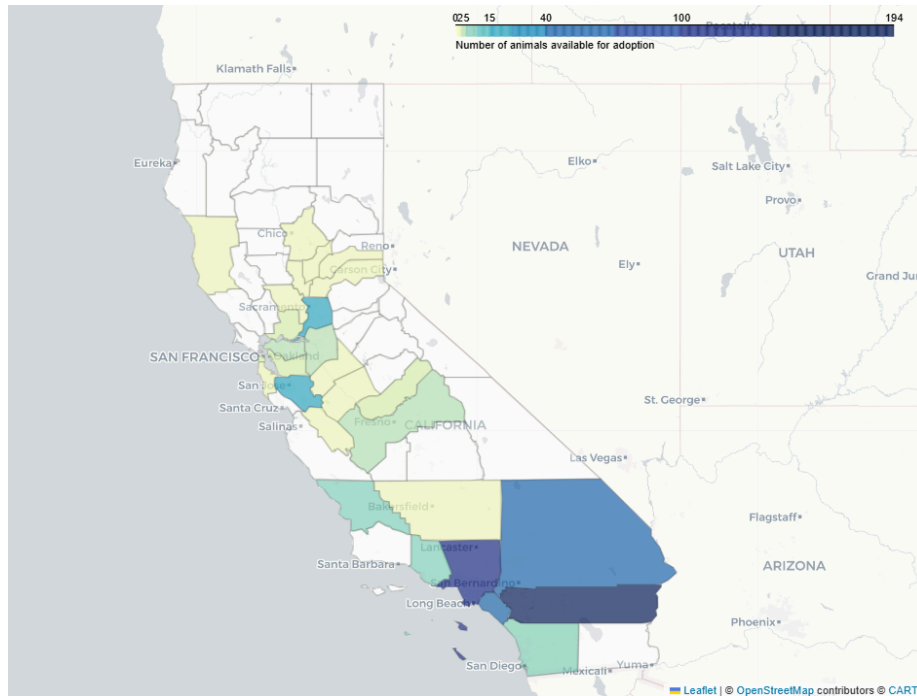


Figure 2: Geospatial Distribution of Rescue Pressure Across California Counties

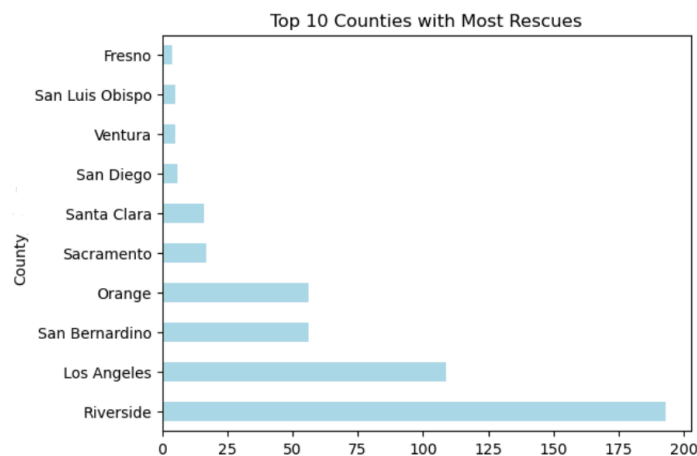


Figure 3: Top 10 California Counties With the Highest Number of Rescues

4.2.2 Breed Diversity and Urgency Identification

As illustrated in the breed distribution donut chart (Figure 4), we implemented a "Top 10 + Others" strategy to manage high-cardinality categorical data. The visualization shows that German Shepherds and their mixes occupy the largest number. The high proportion of German Shepherd mixes indicates that large, high-energy dogs are common in California, which may reflect a higher rescue burden for breeds requiring significant space and exercise. The fact that nearly 60% of pets fall into the "Other Breeds" category highlights an immense diversity, suggesting that adopters have a wide variety of choices. These findings directly support our goal of optimizing the matching mechanism. By clarifying the breed landscape, we help potential adopters understand the diverse options available, potentially increasing the adoption rates for non-primary breeds.

Furthermore, by synthesizing official labels with our keywords, we identified a total of 81 animals currently in an urgent status. The adoption urgency bar chart (Figure 5) reveals that many at-risk animals were previously invisible to standard search filters, as their important information was only manually recorded in their names. Our system ensures that rescue resources are precisely allocated to the pets that are most in need of adoption.

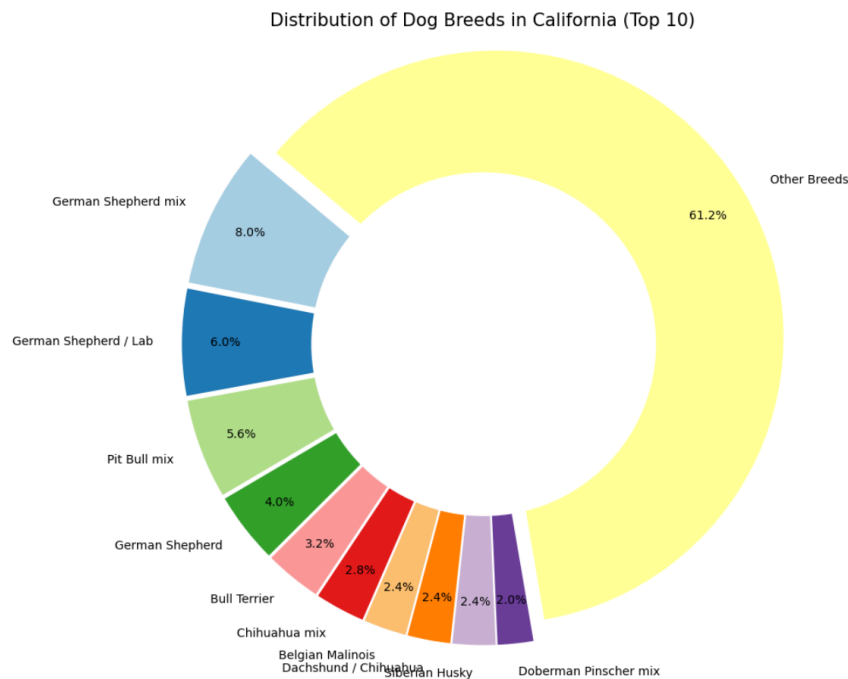


Figure 4: Distribution of Dog Breeds in California Using a "Top 10 + Others" Aggregation Strategy

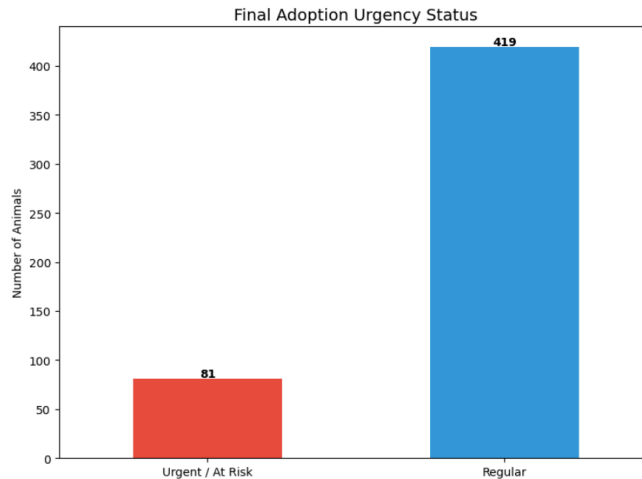


Figure 5: Final Adoption Urgency Status: Combining Official Labels With Keywords

4.2.3 Household Compatibility and Adoption Decision Support

The Stacked Bar Chart (Figure 6) provides a roadmap for potential adopters by modeling the nested relationship of pet compatibility. We compared the compatibility of dogs and cats across three dimensions: "Friendly with Kids," "Friendly with Kids & Dogs," and "Triple Friendly" (Kids, Dogs, and Cats). As family structures become more complex, the number of suitable pets gradually decreases. Our analysis shows a distinct gap between species: dogs demonstrate significantly better performance in adapting to multi-member households compared to cats. However, for households with only children and no other pets, cats remain an excellent choice, with nearly 50% being labeled "kid-friendly." This visualization offers direct value to the adoption process. We empower adopters to identify pets that match their specific family structure, ultimately reducing the likelihood of animals being returned to shelters due to household friction.

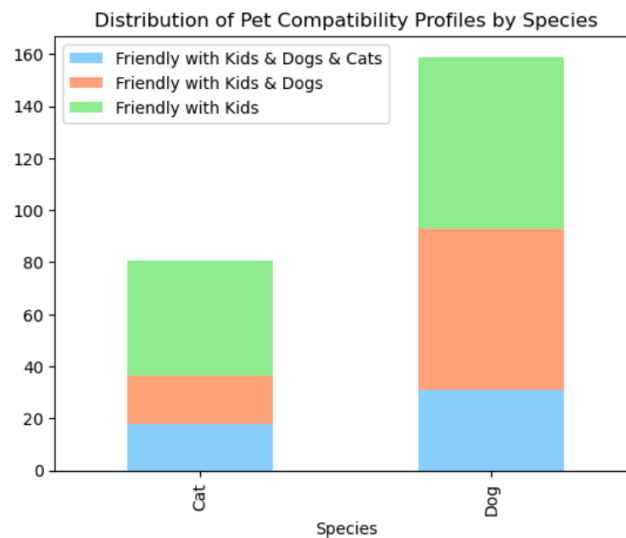


Figure 6: Comparative Analysis of Multi-Level Household Compatibility Profiles by Species

4.2.4 Personality Profiles

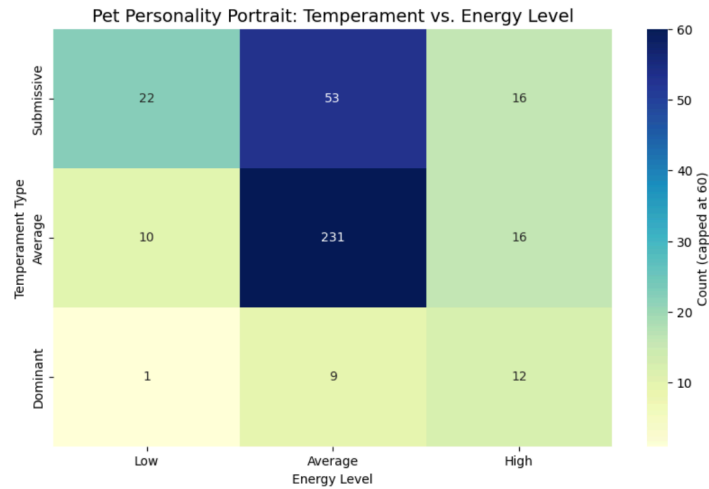


Figure 7: Behavioral Portrait: Heatmap of Pet Temperament vs. Energy Level

The behavioral portrait of the population is captured in the heatmap (Figure 7). Using color capping, we cross-analyzed Temperament Type against Energy Level. The data reveals that the majority of rescue animals possess a balanced behavioral profile, falling into the "Double Average" category. This suggests that the shelter population is more stable and predictable than commonly perceived. A key insight is that animals with a "Submissive" temperament outnumber those with a "Dominant" one, which may ease the concerns of first-time adopters. And the rarest combination is "Dominant and Low Energy", indicating that behavioral extremes are rare in the current population. This behavioral profile helps us manage adopters' expectations and reduce misconceptions. It helps break the stereotype that "shelter pets have behavioral issues," and instead demonstrates that most pets are emotionally stable, ordinary pets suited for standard family environments.

4.2.5 Health Status and Veterinary Welfare Assessment

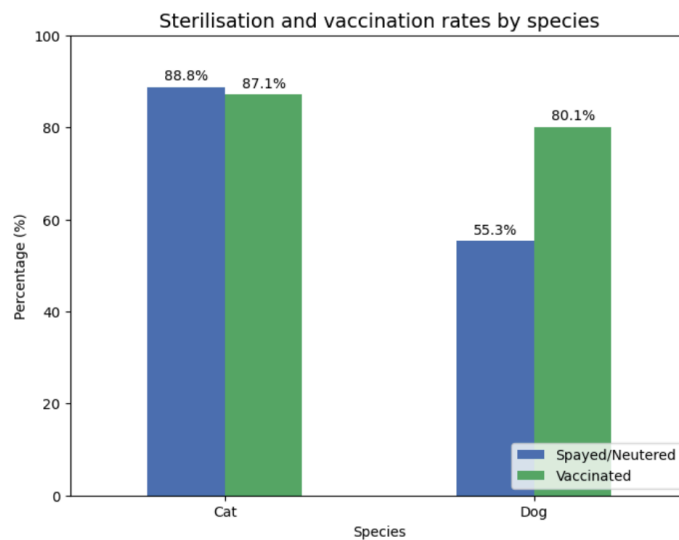


Figure 8: Vaccination and Spay Rates for Cats and Dogs in California

Figure 8 focuses on vaccination and spay rates for both cats and dogs. The grouped bar chart visualizes the percentage of animals that have received these essential medical interventions before being listed for adoption. Both cats and dogs show high vaccination rates (above 80%), indicating strong preliminary medical care. However, a significant discrepancy was found in reproductive health: the neutering rate for dogs is only 56.0%, which is lower than that for cats. This insight suggests that future animal welfare initiatives in California should prioritize canine neutering programs to manage the long-term stray population more effectively.

4.3 Drivers of Adoption Speed

While the previous section focused on descriptive patterns in the dataset, this section investigates the factors that influence how quickly pets are adopted. We analyze both structured variables and textual descriptions to identify key drivers of AdoptionSpeed.

4.3.1 Effects of Structured Variables

Before incorporating text analysis, we first examined the relationship between adoption speed and several structured attributes in the PetFinder dataset. Correlation analysis suggests that certain observable characteristics are associated with adoption outcomes.

Figure 9 presents the correlation matrix between AdoptionSpeed and a set of structured variables, including photo count, age, sterilization status, vaccination status, and other observable attributes. The heatmap suggests that some variables exhibit noticeable associations with adoption outcomes. Since lower values of AdoptionSpeed correspond to faster adoption, variables negatively correlated with AdoptionSpeed indicate factors associated with faster adoption. In particular, pets with more photos tend to be adopted more quickly, while older animals are generally associated with slower adoption speeds. Health-related indicators such as sterilization and vaccination also show moderate correlations with adoption outcomes.

However, correlation alone does not fully describe the explanatory power of these variables. To further evaluate their predictive contribution, we computed the coefficient of determination (R^2) for individual predictors.

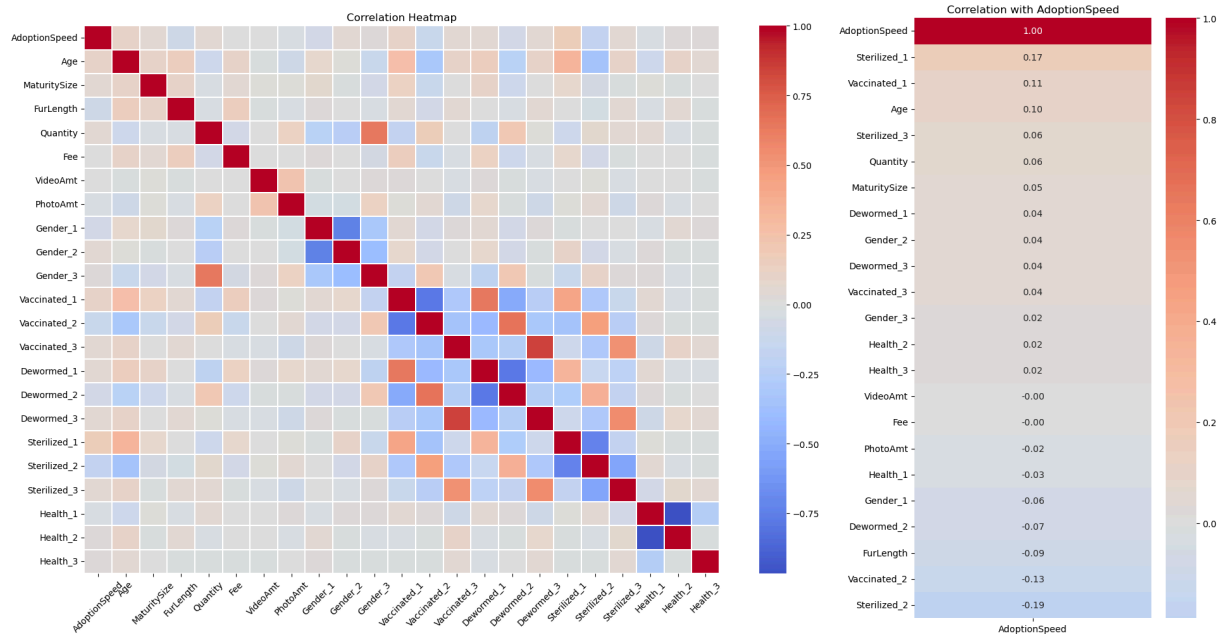


Figure 9. Correlation Between AdoptionSpeed and Key Structured Features

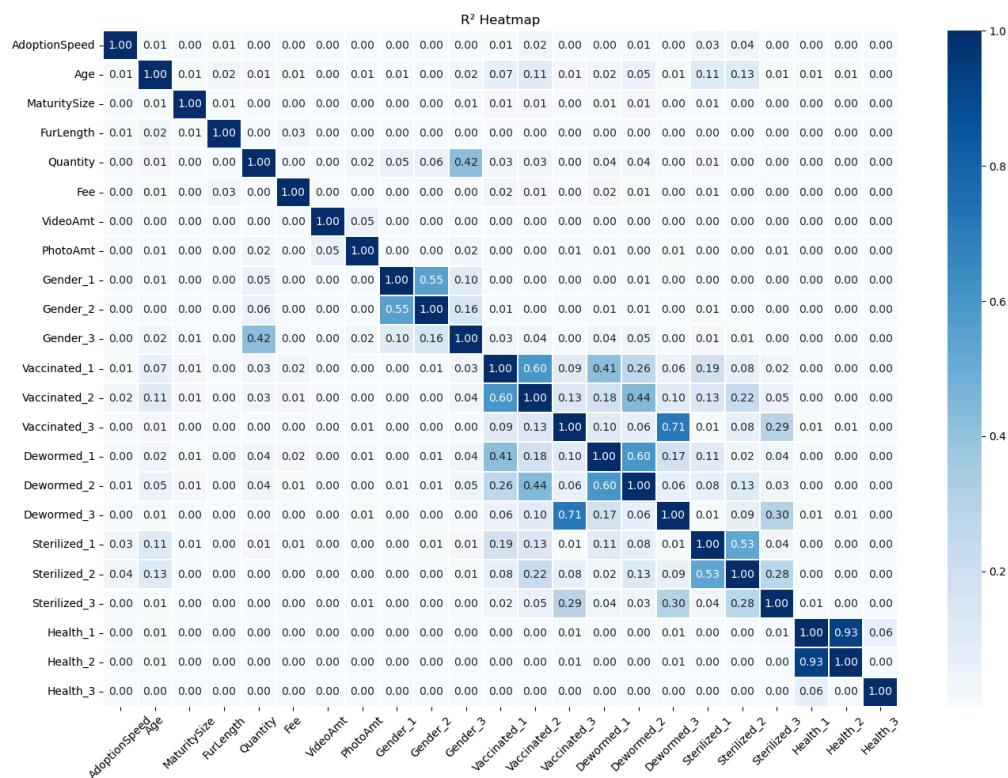


Figure 10. R²: The Explanatory Power of Structured Variables for AdoptionSpeed

As shown in Figure 10, most structured variables explain only a very small proportion of the variance in AdoptionSpeed. In many cases, the R² values are below 5%, indicating that individual attributes provide limited predictive power when considered in isolation.

These findings suggest that while structured variables capture some observable characteristics of pets, they may not fully explain adoption decisions.

4.3.2 NLP Analysis of Pet Descriptions

To further explore the role of textual information in adoption outcomes, we applied Natural Language Processing techniques to the pet descriptions. After tokenization and part-of-speech tagging, adjectives describing personality and behavioral traits were extracted from each description. Using TF-IDF weighting, we selected the top 20 most informative adjectives and incorporated them into an Ordered Logistic Regression model.

	coef	std err	z	P> z	[0.025	0.975]
desc_len	-2.607e-06	4.55e-05	-0.057	0.954	-9.18e-05	8.65e-05
interested	-0.2000	0.040	-4.983	0.000	-0.279	-0.121
good	0.1459	0.039	3.779	0.000	0.070	0.222
puppy	-0.1355	0.045	-2.998	0.003	-0.224	-0.047
active	-0.1247	0.044	-2.861	0.004	-0.210	-0.039
healthy	-0.1242	0.043	-2.912	0.004	-0.208	-0.041
playful	-0.1536	0.039	-3.910	0.000	-0.231	-0.077
adorable	-0.1259	0.053	-2.379	0.017	-0.230	-0.022
little	-0.1421	0.052	-2.750	0.006	-0.243	-0.041
friendly	0.2843	0.045	6.386	0.000	0.197	0.372
free	0.0661	0.067	0.981	0.327	-0.066	0.198
cute	-0.0951	0.045	-2.097	0.036	-0.184	-0.006
sweet	0.0991	0.059	1.670	0.095	-0.017	0.215
beautiful	-0.0317	0.069	-0.458	0.647	-0.168	0.104
happy	0.0030	0.076	0.040	0.968	-0.145	0.151
willing	-0.0798	0.075	-1.065	0.287	-0.227	0.067
small	-0.0725	0.075	-0.960	0.337	-0.220	0.075
able	0.0580	0.074	0.789	0.430	-0.086	0.202
ready	0.0447	0.076	0.587	0.557	-0.105	0.194
contact	0.1426	0.041	3.494	0.000	0.063	0.223
long	-0.2069	0.057	-3.650	0.000	-0.318	-0.096
0/1	-3.6636	0.054	-68.079	0.000	-3.769	-3.558
1/2	0.8713	0.020	43.429	0.000	0.832	0.911
2/3	0.1906	0.014	13.345	0.000	0.163	0.219
3/4	-0.0574	0.016	-3.570	0.000	-0.089	-0.026

Table 2. Estimated Coefficients from the Ordered Logistic Model Using NLP Features

The estimated regression coefficients are summarized in Figure 11, where several adjectives show statistically significant associations with adoption outcomes. Words such as *playful*, *active*, *healthy*, *adorable* and *cute* have negative coefficients, indicating that pets described using these terms are more likely to be adopted quickly. These descriptors often emphasize positive personality traits that may make animals more appealing to potential adopters.

In contrast, commonly used descriptors such as *friendly* and *good* exhibit positive coefficients, suggesting relatively slower adoption speeds. One possible explanation is that these terms appear frequently across many listings and therefore provide limited distinguishing information for potential adopters. As a result, they may contribute less to differentiating highly desirable animals from the rest of the population.

Interestingly, the length of the description does not show a statistically significant effect on adoption speed, indicating that simply providing longer descriptions does not necessarily improve adoption outcomes. Overall, these results highlight the importance of narrative framing in adoption listings and

suggest that the specific wording used in descriptions, rather than their length, can meaningfully influence adoption dynamics.

4.3.3 Adoption Speed Prediction for the California Dataset

After training the Ordered Logistic Regression model on the PetFinder dataset, we developed a prediction function to estimate adoption speed for new pet descriptions. The function first processes each description using the same feature extraction pipeline described earlier, including computing the description length and generating binary indicators for the presence of the selected adjectives.

Based on these features, the trained model produces predicted probabilities for each adoption speed category. The function then computes the expected adoption speed, defined as the probability-weighted average of the predicted categories, to obtain a single predicted score for each pet.

	Name	Species	Breed	predicted_speed
	Nala	Dog	Italian Greyhound / Jack Russell	2.11
	JACK & JAZZY	Cat	Domestic Cat	2.17
	Zoe	Dog	Shih Tzu / Maltese	2.23
	Theo (Courtesy Post)	Dog	Siberian Husky / Pit Bull	2.24
	PETEY	Dog	Bull Terrier / Lab	2.25
	Boo Boo	Dog	Pomeranian mix	2.26
	Uzi	Cat	Domestic Cat	2.29
	Sapphire	Dog	Pit Bull	2.29
	Cuervo	Dog	Belgian Malinois / German Shepherd	2.30
	Bella	Dog	Maltese / Poodle	2.30

Table 3. Top 10 Pets With the Fastest Predicted Adoption Speeds

We applied this prediction function to the California rescue dataset collected through our web scraping pipeline, generating predicted adoption speed scores for all animals in the dataset. Figure 12 shows the top 10 pets with the fastest predicted adoption speeds. These animals typically have descriptions containing positive behavioral traits such as being *playful* or *active*, which are associated with faster adoption in the regression analysis.

This application demonstrates how the trained model and prediction function can be used to estimate adoption outcomes for newly collected rescue listings.

5. Summary

This project shows how structured data analysis and predictive modeling can improve the organization and accessibility of pet adoption information. Listings from [RescueMe.org](https://www.rescueme.org) were scraped to create a dataset describing pets' locations, behavior, compatibility and health traits. The results show that rescue pressure is concentrated in a few California counties, especially Riverside County. Many animals displayed balanced temperament and compatibility traits suitable for typical households. The text-based detection methods identified urgent adoption cases that were missed by the platform's standard filters. The predictive analysis suggests that richer descriptions and more photos are associated with faster adoption.

Several limitations should be noted. First, the dataset is relatively small, consisting of 250 listings each for dogs and cats from California. While this is sufficient for this project, a larger dataset across more states and species would better represent the adoption landscape. Second, many listings contain missing

information on compatibility, health, or temperament. These values were coded as "Unknown" to preserve the observations. Finally, the analysis relies on the accuracy of the original listings, which may sometimes be incomplete or inaccurate.

Future work could expand this project in several ways. One way is to develop a matching interface for adopters, similar to a swipe-style system, where users can browse pets based on traits such as energy level, temperament, and household compatibility. Another direction is to collaborate with local shelters and rescues to help organize their data into structured formats and improve filtering tools on their websites. These improvements could make the adoption process more efficient and help adopters find pets that best match their lifestyles.

7. References

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8. List of Figures and Tables

Figures

- Figure 1: Visual Representation of the Web Scraping Process
- Figure 2: Geospatial Distribution of Rescue Pressure Across California Counties
- Figure 3: Top 10 California Counties With the Highest Number of Rescues
- Figure 4: Distribution of Dog Breeds in California Using a "Top 10 + Others" Aggregation Strategy
- Figure 5: Final Adoption Urgency Status: Combining Official Labels With Keywords
- Figure 6: Comparative Analysis of Multi-Level Household Compatibility Profiles by Species
- Figure 7: Behavioral Portrait: Heatmap of Pet Temperament vs. Energy Level
- Figure 8: Vaccination and Spay Rates for Cats and Dogs in California
- Figure 9: Correlation Between AdoptionSpeed and Key Structured Features
- Figure 10: R^2 : The Explanatory Power of Structured Variables for Adoption Speed

Tables

- Table 1: Data Frame of Rescue Pet Attributes
- Table 2: Estimated Coefficients From the Ordered Logistic Model Using NLP Features
- Table 3: Top 10 Pets With the Fastest Predicted Adoption Speeds